

Ex.no:3	DATA CLEANING

Aim:

To perform data cleaning operations on the **Tourism Dataset** using Python and Pandas. The operations include checking missing values, handling missing values using different techniques, and removing missing values from the dataset.

Algorithm:

Check Missing Values:

1. Start.
2. Import the Pandas library.
3. Load the Tourism dataset into a DataFrame.
4. Display the dataset information using `df.info()`.
5. Check missing values using `df.isnull()`.
6. Count missing values using `df.isnull().sum()`.
7. Check whether any missing values exist using `df.isnull().values.any()`.
8. Check non-missing values using `df.notnull().sum()`.
9. Display the results.
10. Stop.

Fill Missing Values

1. Start.
2. Import the Pandas library.
3. Load the Tourism dataset into a DataFrame.
4. Fill missing values with a constant value using `fillna()`.
5. Fill numerical missing values using the **mean**.
6. Fill numerical missing values using the **median**.
7. Fill categorical missing values using the **mode**.
8. Fill missing values using the **forward fill** method.
9. Fill missing values using the **backward fill** method.
10. Display the results.
11. Stop.

Drop Missing Values

1. Start.
2. Import the Pandas library.
3. Load the Tourism dataset into a DataFrame.
4. Drop rows containing missing values using `dropna()`.
5. Drop rows or columns using the **axis** parameter.

6. Use `how= 'any'` to remove rows containing any missing value.
7. Use `how= 'all'` to remove rows only when all values are missing.
8. Use `thresh=N` to retain rows having at least N non-missing values.
9. Use `subset=[ 'column_name' ]` to drop rows based on specific columns.
10. Use `inplace=True` to permanently modify the DataFrame.
11. Display the cleaned DataFrame.
12. Stop.

Program:

1. Checking Missing Values :

```
import pandas as pd
```

```
# Load Tourism Dataset
```

```
df = pd.read_csv("tourism_dataset_200.csv")
```

```
print("===== DATASET INFO =====")
```

```
df.info()
```

```
print("\n===== MISSING VALUES IN EACH COLUMN =====")
```

```
print(df.isnull().sum())
```

```
print("\n===== TOTAL MISSING VALUES =====")
```

```
print(df.isnull().sum().sum())
```

```
print("\n===== ANY MISSING VALUES? =====")
```

```
print(df.isnull().values.any())
```

```
print("\n===== NON-MISSING VALUES COUNT =====")
```

```
print(df.notnull().sum())
```

2. Filling Missing Values

```
import pandas as pd

# Load Tourism Dataset
df = pd.read_csv("tourism_dataset_200.csv")

# Fill all missing values with 0
df_constant = df.fillna(0)

print("===== CONSTANT VALUE =====")
print(df_constant.head())

# Fill 'Rating' column with mean
df_mean = df.copy()

df_mean["Rating"] = df_mean["Rating"].fillna(
    df_mean["Rating"].mean()
)

print("\n===== MEAN =====")
print(df_mean.head())

# Fill 'Average_Duration_Days' column with median
df_median = df.copy()

df_median["Average_Duration_Days"] = \
    df_median["Average_Duration_Days"].fillna(
        df_median["Average_Duration_Days"].median()
    )

print("\n===== MEDIAN =====")
```

```
print(df_median.head())
```

```
# Fill 'Country' column with mode
```

```
df_mode = df.copy()
```

```
df_mode["Country"] = df_mode["Country"].fillna(  
    df_mode["Country"].mode()[0]  
)
```

```
print("\n===== MODE =====")
```

```
print(df_mode.head())
```

```
# Forward Fill
```

```
df_ffill = df.ffill()
```

```
print("\n===== FORWARD FILL =====")
```

```
print(df_ffill.head())
```

```
# Backward Fill
```

```
df_bfill = df.bfill()
```

```
print("\n===== BACKWARD FILL =====")
```

```
print(df_bfill.head())
```

3. Dropping Missing Values :

```
import pandas as pd
```

```
# Load Tourism Dataset
```

```
df = pd.read_csv("tourism_dataset_200.csv")
```

```
# Drop rows with missing values
```

```
df_drop_rows = df.dropna()
```

```
print("===== DROP ROWS =====")
```

```
print(df_drop_rows.head())
```

```
# Drop columns with missing values
```

```
df_drop_cols = df.dropna(axis=1)
```

```
print("\n===== DROP COLUMNS =====")
```

```
print(df_drop_cols.head())
```

```
# Drop rows having any missing value
```

```
df_any = df.dropna(how="any")
```

```
print("\n===== DROP HOW='ANY' =====")
```

```
print(df_any.head())
```

```
# Drop rows where all values are missing
```

```
df_all = df.dropna(how="all")
```

```
print("\n===== DROP HOW='ALL' =====")
```

```
print(df_all.head())
```

```
# Keep rows with at least 10 non-null values
```

```
df_thresh = df.dropna(thresh=10)

print("\n===== THRESH=10 =====")
print(df_thresh.head())

# Drop rows where 'Country' or 'Rating' is missing
df_subset = df.dropna(
    subset=["Country", "Rating"]
)

print("\n===== SUBSET (Country, Rating) =====")
print(df_subset.head())

# Drop missing values permanently
df.dropna(inplace=True)

print("\n===== AFTER inplace=True =====")
print(df.head())
```

Output:

===== DATASET INFORMATION =====

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Tourism_ID	200 non-null	object
1	Destination	200 non-null	object
2	Country	200 non-null	object
3	Continent	200 non-null	object
4	Tourism_Type	200 non-null	object
5	Best_Season	200 non-null	object
6	Average_Duration_Days	200 non-null	int64
7	Estimated_Budget_USD	200 non-null	int64
8	Rating	200 non-null	float64
9	Annual_Visitors	200 non-null	int64
10	Accommodation_Type	200 non-null	object
11	Transportation	200 non-null	object

dtypes: float64(1), int64(3), object(8)

memory usage: 18.9+ KB

===== FIRST 5 RECORDS =====

	Tourism_ID	Destination	Country	Continent	Tourism_Type	Best_Season	\
0	T001	Los Angeles	USA	North America	City	Winter	
1	T002	Barcelona	Spain	Europe	City	Winter	
2	T003	Paris	France	Europe	Wildlife	Winter	
3	T004	Singapore	Singapore	Asia	Wildlife	Spring	
4	T005	Tokyo	Japan	Asia	Cultural	Spring	

	Average_Duration_Days	Estimated_Budget_USD	Rating	Annual_Visitors	\
0	3	504	4.6	133393	
1	11	3756	3.5	54123	
2	10	1928	4.6	372696	
3	14	353	4.6	88707	
4	14	3057	3.7	204191	

	Accommodation_Type	Transportation
0	Hotel	Cruise
1	Apartment	Cruise
2	Homestay	Cruise
3	Hostel	Train
4	Hostel	Cruise

===== MISSING VALUES IN EACH COLUMN =====

Tourism_ID	0
Destination	0
Country	0
Continent	0
Tourism_Type	0
Best Season	0

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Tourism_ID                            200 non-null    object
1   Destination                            200 non-null    object
2   Country                               200 non-null    object
3   Continent                             200 non-null    object
4   Tourism_Type                           200 non-null    object
5   Best_Season                           200 non-null    object
6   Average_Duration_Days                 200 non-null    int64
7   Estimated_Budget_USD                 200 non-null    int64
8   Rating                                200 non-null    float64
9   Annual_Visitors                       200 non-null    int64
10  Accommodation_Type                    200 non-null    object
11  Transportation                         200 non-null    object
dtypes: float64(1), int64(3), object(8)
memory usage: 18.9+ KB

===== DATA CLEANING COMPLETED SUCCESSFULLY =====

```

Result: